Hareen Maddipatla

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SUMMARY

Data Scientist with experience modeling **1M+ records** across public safety, energy, and mobility datasets, delivering **75% classification accuracy** on imbalanced problems. Built **Bayesian probabilistic models** (NumPyro/JAX) and **time-series forecasting pipelines** (SARIMA) with rigorous diagnostics, calibration, and out-of-sample validation. Strong at turning noisy, real-world data into clean pipelines and decision-ready outputs.

SKILLS

- **Programming Languages**: Python, R, SAS, SQL
- **Machine Learning:** Scikit-learn, XGBoost, Random Forest, K-Means, Gaussian Mixture Models, Regression, Classification, Model Evaluation
- **Statistics & Forecasting:** Bayesian Modeling, Time Series Forecasting, Hypothesis Testing, MANOVA, Experimental Design
- **Analytics & Visualization:** Power BI, Tableau, Plotly Dash, Matplotlib, Seaborn, Excel
- **Data Workflow:** Data Cleaning, Feature Engineering, Missing Value Treatment, Outlier Analysis, Git, GitHub, Jupyter Notebook, VS Code

EDUCATION

University of Texas at Arlington — **Arlington, Texas**
Master of Science (M.S) in Applied Statistics and Data Science (CGPA: 4.0) — **Aug 2024 – May 2026**

- Relevant Courses: Statistical Theory, Statistical Computing, Data Visualization, Time Series Analysis, Multivariate Analysis.

WORK EXPERIENCE

University of Texas at Arlington — **Arlington, Texas**
Teaching Assistant — ***Jan 2025 – May 2026***

- Supported 150+ students across four data and statistics courses by redesigning Python and statistics walkthroughs, improving assignment completion quality from inconsistent submissions to consistently correct implementations.
- Introduced GitHub Classroom to replace manual grading workflows, reducing average grading turnaround time from ~5 days to ~4 days (20% improvement) and improving submission traceability and version control.
- Systematically analyzed student feedback and office-hour attendance trends, refining tutorial structure and scheduling, resulting in higher engagement consistency and fewer repeat clarification requests.

Tek Certify Training Solutions Pvt Ltd — **Hyderabad, Telangana**
Technology Consultant — ***Aug 2023 – July 2024***

- Delivered cybersecurity and AI training to 70+ candidates by redesigning lab-based modules and case workflows, shifting sessions from lecture-heavy delivery to hands-on, code-driven learning.
- Analyzed structured client feedback (5 recurring issues) to re-engineer curriculum sequencing, reducing revision cycle time from ad-hoc updates to a structured process with a 20% faster turnaround.
- Improved trainee comprehension from approximately 40% to 66% by decomposing complex cybersecurity and AI concepts into modular Jupyter Notebook–based exercises with executable examples.
- Increased overall client satisfaction from baseline levels to +40% by implementing 8 targeted, data-backed curriculum and delivery improvements derived from feedback analysis.

SmartKnower — **Hyderabad, Telangana**
Artificial Intelligence Intern — ***May 2023 – Aug 2023***

- Built and evaluated deep learning models for hand-sign recognition, iterating on preprocessing and model tuning to reach >94% accuracy.
- Analyzed training errors and failure cases to identify choke points in the pipeline (data quality / feature patterns), then applied targeted fixes to improve prediction stability.
- Compared candidate DL approaches and selected the best-performing model based on validation results and error analysis, not assumptions or “one-model” bias.
- Documented the end-to-end workflow (data -> training -> evaluation) to make results reproducible and easy to hand off.

PROJECT EXPERIENCE

University of Texas at Arlington — **Arlington, Texas**
Signals to Structural Insight: A Multimodal Framework for Damage Detection — ***Jan 2026 - May 2026***

- Developed an end-to-end multimodal damage analysis pipeline for GFRP composite failure monitoring, integrating acoustic emission, DIC, thermal imaging, strain plots, and timestamp synchronization into a unified Python workflow.
- Engineered acoustic emission features including cumulative energy, cumulative hits, hit rate, rolling energy, rolling amplitude, and normalized time to capture damage accumulation, instability bursts, and failure stage transitions.
- Applied Gaussian Mixture Models to segment acoustic emission events into interpretable damage regimes, enabling unsupervised identification of quiet response, microdamage, progressive damage, and failure onset behavior.

- Built an interactive Plotly Dash dashboard to visualize synchronized AE trends, ML cluster outputs, DIC frames, thermal images, and strain evolution, improving interpretability of multimodal composite damage progression.

Understanding Team Strength Through Bayesian Soccer Modelling ***Nov 2025 - Dec 2025***

- Built a hierarchical Bayesian Dixon-Coles goal model to learn team attack/defense, home advantage, and low-score dependence.
- Cleaned and encoded 12 EPL seasons (2012–2024; ~3.4K matches), added time-decay weights, and held out 2024–25 for forward testing.
- Ran vectorized NUTS (HMC), enforced sum-to-zero identifiability constraints, and validated convergence via R-hat/ESS/trace diagnostics.
- Generated Home/Draw/Away probabilities via posterior predictive simulation; evaluated with RPS + calibration vs bookmaker baseline (test RPS $\approx$ 0.230).

Understanding and Predicting Bike Rental Behavior ***Nov 2025 - Dec 2025***

- Built a forecasting pipeline on 2 years of bike-share demand (daily + hourly), structuring time-indexed series and separating trend, seasonality, and noise.
- Validated stationarity assumptions using ADF tests plus ACF/PACF and spectral analysis, then applied first-order and seasonal differencing to stabilize variance.
- Fit and selected SARIMA models via parallelized grid search using AIC, then stress-checked residual diagnostics to avoid overfitting.
- Produced 7-day hourly forecasts that preserved the 24-hour seasonal cycle; reported MAE/RMSE and compared predicted vs observed demand patterns

A Multivariate Approach to Differentiate Benign and Malignant Breast Tumors ***Nov 2025 - Dec 2025***

- Analyzed the Wisconsin Diagnostic Breast Cancer (WDBC) dataset (569 cases, 30 features) by structuring correlated nuclear morphology variables into multivariate response blocks and validating data integrity prior to modeling.
- Applied one-way, repeated-measures, and two-way MANOVA using Pillai's trace to robustly test benign–malignant separation under unequal covariance structures, confirming strong multivariate differentiation (Pillai $\approx$ 0.68, $p < 0.0001$).
- Used canonical discriminant analysis and Hotelling's T^2 to isolate clinically interpretable signals, showing malignant tumors exhibit jointly larger nuclei and greater border irregularity, with separation strengthening in larger-nucleus strata.

From Minutes to Meaning: Clustering and Classifying Energy Behavior ***Mar 2025 - Apr 2025***

- Transformed high-frequency minute-level household electricity data (2006–2010) into a time-aware, model-ready dataset by handling missing values via interpolation, aggregating to hourly resolution, and standardizing features to preserve behavioral and seasonal patterns.
- Applied unsupervised clustering (K-Means) to segment residential energy usage into distinct behavioral profiles (low, moderate, peak load), then operationalized these clusters using KNN and C5.0 decision tree classifiers to enable consistent classification of new observations.
- Achieved 98%+ classification accuracy (KNN: 98.65%, C5.0: 98.12%, Kappa > 0.96), validating cluster separability and combining high predictive performance with interpretable, rule-based decision paths for real-world energy analytics.

Mapping & Modelling Crime in Los Angeles with Geo-Analytics ***Mar 2025 - Apr 2025***

- Processed and engineered LAPD crime records (2020–present), resolving missing data, encoding high-cardinality categoricals, and handling outliers to create a model-ready dataset.
- Conducted spatial-temporal EDA to identify hotspots, peak hours, and low arrest-rate patterns, informing feature design and target definitions.
- Trained and tuned Random Forest and XGBoost classifiers with SMOTE/re-sampling to address severe class imbalance; evaluated via accuracy and weighted F1.
- Achieved up to 75% accuracy for case-status prediction and translated results into actionable patrol and resource-allocation insights.

Song Popularity Prediction ***Mar 2025 - Apr 2025***

- Analyzed a dataset of 18,835 songs with 15 audio features by performing structured EDA, correlation analysis, and visualization to identify statistically relevant drivers of song popularity.
- Built and evaluated multiple regression models (Multiple Linear, Ridge, Lasso, Elastic Net, Polynomial) after preprocessing steps including duplicate removal, IQR-based outlier handling, one-hot encoding, feature scaling, and PCA-based dimensionality reduction.
- Quantified model limitations using R^2, RMSE, and MAE (best $R^2 \approx 0.019$), and supplemented weak predictive performance with hypothesis testing (t-tests, ANOVA, Kruskal-Wallis, Pearson correlation) to extract statistically valid business insights.

Osmania University **Hyderabad, Telangana**

Cancer and Deaths Data Analysis with Power BI ***Mar 2024 - Apr 2024***

- In my work on cancer research, I developed a suite of interactive Power BI dashboards to analyze global trends in cancer and death data spanning 1990 to 2019. Leveraging data from Kaggle's "Cancer and Deaths Dataset (1990-2019) Globally," these dashboards provide insights into various aspects of the data, including global deaths by year with a map view, deaths by age group with a focus on specific ranges, the number of people affected by cancer across different age groups over time, comparisons of cancer cases in specific age groups, and multi-year cancer analysis for each age category. This project demonstrates my ability to analyze and visualize complex datasets using Power BI, contributing to a greater understanding of global cancer trends.

Data Analysis and Prediction of Football Goals ***Oct 2022 - May 2023***

- Sports enthusiasts exhibit a profound interest in even the minutest aspects of a game, and their access to a wealth of statistical information renders them more knowledgeable. This discourse delves into how advanced analytics and predictive modeling surpass traditional analysis methods in the realm of sports. The application of Artificial Intelligence and state-of-the-art Machine Learning models have already demonstrated superior efficacy compared to archaic analyst techniques, transforming various industries. Despite the profusion of available data, the integration of analytics in sports has been erratic. In this project, a dataset is created by converting sample videos into analyzable data. The conversion process entails video tracking utilizing OpenCV, while the project's programming language shall be Python. The aim is to showcase how cutting-edge analytical approaches can offer insights beyond traditional analysis and foster a more comprehensive comprehension of the sports world.

LEADERSHIP EXPERIENCE

IEEE, MVSREC **Hyderabad, Telangana**

Computer Society Chairperson **Aug 2022 – Jun 2023**

- Planned and delivered 7 IEEE events, coordinating execution across a 20-member team and multi-channel outreach to drive 300+ attendees.
- Built data-driven marketing campaigns for Technovanza-2K23 using Google Analytics, increasing social engagement by 10% and driving 900+ participants.
- Analyzed attendee feedback in Excel and translated insights into event redesigns (agenda/logistics), improving satisfaction scores by 60%.
- Standardized event KPI tracking (turnout, engagement, satisfaction) to iterate faster across events and improve repeatable execution quality.